

Subject Description Form

Subject Code	ME49006
Subject Title	Integrated Capstone Project
Credit Value	6
Level	4
Pre-requisite / Co-requisite/ Exclusion	Pre-requisite: ME31001 Dynamics and Vibrations; ME31002 Linear Systems and Control; ME32001 Manufacturing Fundamentals; ME33001 Mechanics of Materials; ME34002 Engineering Thermodynamics; ME34004 Fluid Mechanics; ENG2002 Computer Programming; and ME46002 Numerical Methods for Engineers. Exclusion: ME49001 Final Year Capstone Project
Objectives	The objectives of this subject are to: (a) Provide students with an opportunity of integrating and applying knowledge from different disciplines of mechanical engineering (ME) with artificial intelligence and data analytics (AIDA) to conduct an engineering project that is open-ended and requires team collaboration for its completion; (b) Develop the capabilities of a student in analyzing and solving complex and possibly real-life problems using ME and AIDA knowledge; (c) Train students with skills on systematic development and documentation of a significant piece of work.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: <u>Professional/academic knowledge and skills</u> (a) Conduct literature search including patents, books, archived publications and product catalogues, and to perform the state-of-the-art and benchmark studies; (b) Formulate the problem precisely, and utilize knowledge and skills from ME and AIDA to generate good solutions to the open-ended real-world problems encountered in the project; (c) Think critically the formulation of alternative models and solutions to the problem, in the analysis of approaches to the solution and their implementation; (d) Design, plan and carry out scientific and engineering experiments (physical tests and/or computer numerical simulations) to prove the feasibility of their designed solutions; (e) Design the test apparatus, rigs, assemblies or systems as required by the project; (f) Apply appropriate engineering tool (analytical, experimental, and/or computational) for carrying out tasks in the development and implementation of a designed solution; (g) Select and employ the appropriate manufacturing methods in the production and fabrication of components and assemblies required by the project; <u>Attributes for all-roundedness</u> (h) Work in a professional manner and comply with all applicable standards and regulations in conducting the project; (i) Take into account of safety, legal, environmental protection considerations in an engineering project

	(j) Participate and lead in a multi-functional team, and communicate their project work to sponsors (if any), supervisors, other peer teams, and even non-technical audience and articulate the results and findings with scientific and logical arguments; (k) Evaluate the potential impact of their designed solution on performance, safety, cost and environment, and evaluate the final outcome in an objective manner; (l) Improve presentation and communication skills via oral presentation, and enhance technical report writing skills with proper organization of materials.																																																																																																
Subject Synopsis/ Indicative Syllabus	1. In-depth Study of a Topic Typically Proposed by the Supervisor 2. Project Meeting and Planning 3. Proposal Writing 4. Regular Progress Checking and Reporting 5. Project Documentation 6. Presentation and Demonstration Students are expected to identify a project topic with a supervisor with artificial intelligence and data analytics expertise in ME Department. Students need to demonstrate their knowledge in both ME and AIDA in the project, receiving advice from supervisor. The project should represent requisite effort in analysing and interpreting the data/information obtained, using the principles and techniques learnt from various related subjects. Students are also expected to demonstrate significant analytical and, preferably, research ability in the chosen application domain.																																																																																																
Teaching/Learning Methodology	The Integrated Capstone Project spans across the academic year for two consecutive semesters. The teaching/learning activities include regular project meetings with the supervisor, guided study of project materials, independent project development work and other project management tasks. <table><tr><td>Teaching/Learning Methodology</td><td colspan="12">Outcomes</td></tr><tr><td></td><td>a</td><td>b</td><td>c</td><td>d</td><td>e</td><td>f</td><td>g</td><td>h</td><td>i</td><td>j</td><td>k</td><td>l</td></tr><tr><td>Regular project meetings and guided study</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr></table>	Teaching/Learning Methodology	Outcomes													a	b	c	d	e	f	g	h	i	j	k	l	Regular project meetings and guided study	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓																																																									
Teaching/Learning Methodology	Outcomes																																																																																																
	a	b	c	d	e	f	g	h	i	j	k	l																																																																																					
Regular project meetings and guided study	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓																																																																																					
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><td rowspan="2">Specific assessment methods/tasks</td><td rowspan="2">% weighting</td><td colspan="12">Intended subject learning outcomes to be assessed (Please tick as appropriate)</td></tr><tr><td>a</td><td>b</td><td>c</td><td>d</td><td>e</td><td>f</td><td>g</td><td>h</td><td>i</td><td>j</td><td>k</td><td>l</td></tr><tr><td>Continuous monitoring</td><td>15%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Interim report</td><td>10%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Final report</td><td>50%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Oral examination</td><td>25%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Total</td><td>100%</td><td colspan="12"></td></tr></table> The Integrated Capstone Project will be accessed by the supervisor and other assessors. Attributes to be assessed include, but not limited to, Problem Identification, Problem Solving, Communication and Presentation, Project Management, and Self-Discipline. Integrated Capstone Projects should be problem-oriented and there are no restrictions on the nature of the problem except that it should be relevant to ME and AIDA. The project could be practical, academic or a hybrid in which the student is encouraged but not constrained to have some original contributions.	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)												a	b	c	d	e	f	g	h	i	j	k	l	Continuous monitoring	15%	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	Interim report	10%	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	Final report	50%	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	Oral examination	25%	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	Total	100%												
Specific assessment methods/tasks	% weighting			Intended subject learning outcomes to be assessed (Please tick as appropriate)																																																																																													
		a	b	c	d	e	f	g	h	i	j	k	l																																																																																				
Continuous monitoring	15%	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓																																																																																				
Interim report	10%	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓																																																																																				
Final report	50%	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓																																																																																				
Oral examination	25%	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓																																																																																				
Total	100%																																																																																																

	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Overall Assessment: 1.0 × Continuous Assessment (a) Performance of each student should be assessed individually together with the team’s overall performance by the supervisor, an independent assessor, the peers and an examination panel consisting of at least three academic staff; (b) The continuous monitoring of a project group as a whole and that of each group member on an individual basis are conducted by the supervisor. The interim report is assessed by the independent assessor. The final report is assessed by both the supervisor and the independent assessor. As part of the assessment process, each group member is required to specify his/her own contribution in completing the project when compared to his/her team mates (peer assessment). In case of an industrial-based project, comments will be invited from the industrial supervisor but he/she will not be required to perform the formal assessment; (c) The supervisor monitors and assesses the overall and individual progresses through regular meetings. The interim report should be submitted to the independent assessor at around week 8 of the first semester. The final report submitted before the end-of-year examination is assessed by both the supervisor and the independent assessor. Due consideration of each student’s individual contribution and performance will be taken into account; (d) During the oral examination, every group member is required to present the project especially on his/her significant contribution to the whole project, and respond to the questions addressed to him/her by the examination panel. Marks for oral examination are awarded to individual student by taking into account the group’s overall performance; (e) The assessment system is summarized as shown in the following table: <table><tr><th rowspan="2">Assessor</th><th colspan="5">Assessment Component (% of the total)</th></tr><tr><th>Continuous Monitoring (15)</th><th>Interim Report (10)</th><th>Final Report (25)</th><th>Final Report (25)</th><th>Oral Examination (25)</th></tr><tr><td>Supervisor</td><td>✓</td><td></td><td>✓</td><td></td><td></td></tr><tr><td>Independent Assessor</td><td></td><td>✓</td><td></td><td>✓</td><td></td></tr><tr><td>Examination Panel</td><td></td><td></td><td></td><td></td><td>✓</td></tr></table>	Assessor	Assessment Component (% of the total)					Continuous Monitoring (15)	Interim Report (10)	Final Report (25)	Final Report (25)	Oral Examination (25)	Supervisor	✓		✓			Independent Assessor		✓		✓		Examination Panel					✓
Assessor	Assessment Component (% of the total)																													
	Continuous Monitoring (15)	Interim Report (10)	Final Report (25)	Final Report (25)	Oral Examination (25)																									
Supervisor	✓		✓																											
Independent Assessor		✓		✓																										
Examination Panel					✓																									
Student Study Effort Expected	Class contact:																													
	• Guided study			26 Hrs.																										
	Other student study effort:																													
	• Conducting project			154 Hrs.																										
	• Literature search and private study			72 Hrs.																										
	Total student study effort			252 Hrs																										
Reading List and References	To be advised by supervisor.																													